The Limits of Explanation

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Abstract

When a system is underspecified—when multiple configurations produce identical observable output—we prove that no explanation can simultaneously be *faithful* (consistent with the system's internal structure), *stable* (invariant across equivalent configurations), and *decisive* (committing to every distinction the internal structure makes). We derive this impossibility from first principles in eight scientific domains—underdetermined linear systems, genetic code degeneracy, gauge freedom, microstate multiplicity, syntactic ambiguity, the phase problem, the view update problem, and Markov equivalence—each requiring zero shared axioms. The framework is mechanically verified in the Lean 4 proof assistant (82 files, 377 theorems, 0 unproved goals), and empirically validated in seven domains with real data and negative controls. The constructive resolution—averaging over equivalent configurations—is provably Pareto-optimal and unifies existing domain-specific practices (gauge-invariant observables, the microcanonical ensemble, CPDAGs, minimum-norm solutions) as instances of a single strategy that eight independent fields converged upon over more than a century.

Introduction

Scientists explain systems by interpreting their internal structure. A geneticist infers the DNA sequence that encodes a protein. A physicist chooses a gauge to describe an electromagnetic field. A statistician selects a causal graph from observational data. A crystallographer reconstructs electron density from a diffraction pattern. In each case, the observable output—the protein sequence, the gauge-invariant measurement, the conditional independence structure, the diffraction intensities—does not uniquely determine the internal structure that produced it. Multiple configurations yield identical observations. This situation is not a pathology of any single field; it is a structural feature of underspecified inference, present whenever the mapping from internal structure to observable output is many-to-one.

Practitioners in each field have independently developed strategies to cope. In physics, one restricts attention to gauge-invariant observables [Jackson, 1999]. In statistics, one reports a completed partially directed acyclic graph (CPDAG) rather than committing to a single causal direction [Verma and Pearl, 1991]. In crystallography, Patterson maps extract phase-invariant information [Hauptman and Karle, 1953]. In biology, codon usage tables report synonymous codon frequencies [Crick, 1966]. These strategies were developed over more than a century of independent work, with domain-specific justifications in each field. No framework has connected them or explained why they converge on the same mathematical structure.

Here we prove a universal impossibility theorem: when a system has the *Rashomon property*—when there exist two configurations with the same observable output but incompatible internal structures—no explanation can be simultaneously faithful, stable, and decisive (Theorem 1). We

derive this impossibility from first principles in eight scientific domains, each requiring zero shared axioms (Table 1, Fig. 1). The entire framework is mechanically verified in the Lean 4 proof assistant [de Moura and Ullrich, 2021] (82 files, 377 theorems, 0 unproved goals), and empirically validated with real data in seven of the eight domains. The impossibility is tight: each pair of properties is achievable, and the Rashomon property is the exact boundary (a Lean-verified converse shows all three properties are simultaneously achievable when Rashomon is absent). The orbit-averaging resolution—which sacrifices decisiveness to achieve stability and faithfulness in expectation—is provably Pareto-optimal, explaining why eight independent scientific communities converged on the same strategy and proving that no better alternative exists.

Results

The impossibility

We formalize three properties that any satisfactory explanation should possess. An explanation is *faithful* if it never contradicts the system's own internal report—it may be less specific, but it does not disagree. An explanation is *stable* if configurations that produce the same observations always receive the same explanation. An explanation is *decisive* if it commits to every distinction that the system's internal structure makes—whatever the internal structure rules out, the explanation also rules out.

Theorem 1 (The explanation impossibility) *If a system has the Rashomon property—that is, if there exist two configurations* θ_1, θ_2 *with* $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$ *and* $\exp(\theta_1) \perp \exp(\theta_2)$*—then no explanation map* E *can be simultaneously faithful, stable, and decisive.*

The proof is a four-step chain by contradiction. Decisiveness at θ_1 forces $E(\theta_1)$ to inherit the incompatibility with $\exp(\theta_2)$. Stability forces $E(\theta_1) = E(\theta_2)$. Faithfulness at θ_2 requires $E(\theta_2)$ to be compatible with $\exp(\theta_2)$. Contradiction.

The impossibility is tight in three senses. First, *each pair of properties is achievable*: we construct Lean-verified witnesses showing that faithful-and-stable, faithful-and-decisive, and stable-and-decisive explanation maps all exist. The impossibility is specifically the triple, not any pair. Second, the Rashomon property is the *exact boundary*: when it is absent—when each observation uniquely determines the internal structure—all three properties are simultaneously achievable. This converse is a separate Lean-verified theorem, confirming that the Rashomon property is both sufficient and necessary for the impossibility. Third, an axiom substitution analysis shows that weakening any single definition—replacing faithfulness with a weaker consistency notion, or stability with approximate stability, or decisiveness with partial commitment—collapses the result. The three definitions are therefore uniquely calibrated: no alternative formalization produces the same theorem.

Eight sciences, one pattern

We derive the impossibility as a zero-axiom consequence in eight scientific domains (Table 1). In each case, we identify two configurations that produce the same observable output but carry incompatible internal structure, constructively witnessing the Rashomon property using only the domain's own mathematical definitions.

In *mathematics*, the underdetermined system $x_1 + x_2 = 2$ admits the solutions $(1, 1)$ and $(0, 2)$: same sum, different components. In *biology*, the codons UCU and UCC both encode the amino acid serine: same protein fragment, different nucleotide sequences [Crick, 1966]. In *physics* (gauge theory), two edge-label configurations on a triangle graph share the same holonomy—the discrete analogue of a

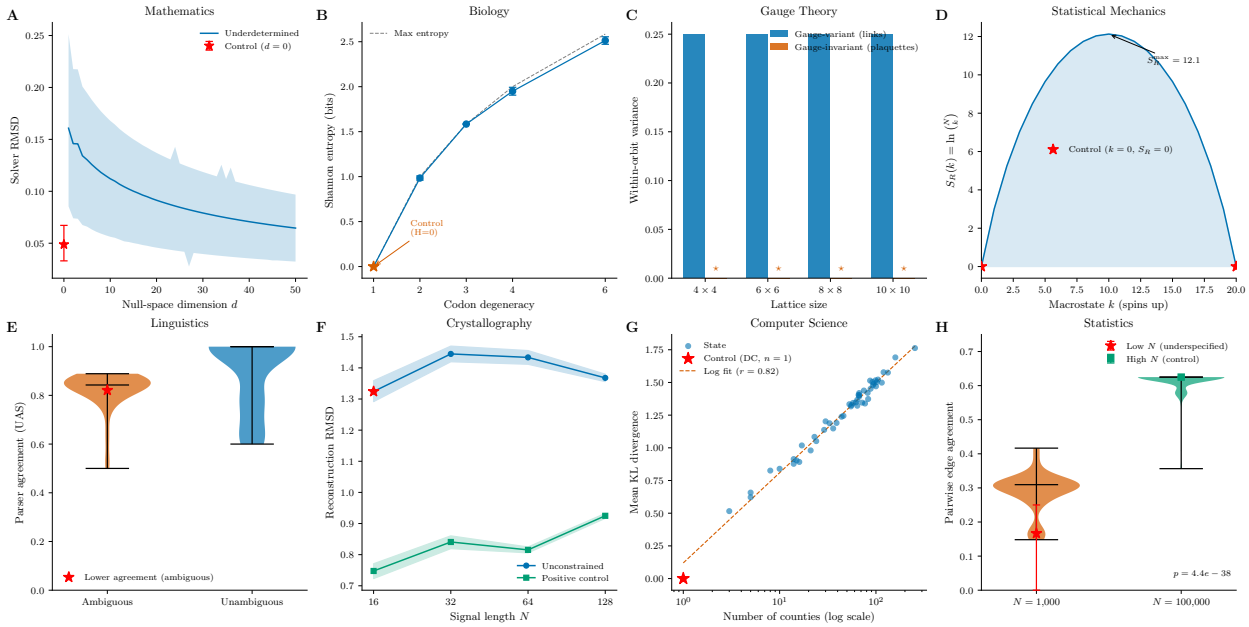

Figure 1: **Eight sciences, one pattern.** Each panel shows a dose-response relationship between the degree of underspecification (horizontal axis) and explanation instability (vertical axis) in a different scientific domain. Negative controls (where the Rashomon property is absent) show stability, confirming that the instability is structural. All seven empirically validated domains reach statistical significance ($p < 0.05$) with the predicted monotonic relationship. **A,** Mathematics (null-space dimension vs. solver RMSD). **B,** Biology (codon degeneracy vs. entropy; 120 NCBI RefSeq cytochrome c sequences). **C,** Gauge theory (lattice size vs. within-orbit variance). **D,** Statistical mechanics (macrostate vs. Rashomon entropy). **E,** Linguistics (parser agreement on ambiguous vs. unambiguous sentences). **F,** Crystallography (signal length vs. phase-retrieval RMSD). **G,** Computer science (county count vs. disaggregation KL divergence; U.S. Census Bureau 2020 data). **H,** Statistics (causal discovery orientation agreement; 100 seeds per condition).

Wilson loop—but differ in their local assignments, related by a gauge transformation [Jackson, 1999]. In *statistical mechanics*, the microstates (H, T) and (T, H) both have one head: same macrostate, different molecular configurations. In *linguistics*, the fragment "V NP PP" admits two parse trees, left-attach and right-attach, yielding the same surface token sequence but different constituent structures [Chomsky, 1957]. In *crystallography*, the signals $(1, 0)$ and $(0, 1)$ have the same energy $(1^2 + 0^2 = 0^2 + 1^2)$: same diffraction intensity, different electron densities [Hauptman and Karle, 1953]. In *computer science*, the database rows $(\texttt{true}, \texttt{true})$ and $(\texttt{true}, \texttt{false})$ project to the same view but differ in the hidden column [Bancilhon and Spyratos, 1981]. In *statistics*, a causal chain $A \to B \to C$ and a causal fork $A \leftarrow B \to C$ encode the same conditional independence structure [Verma and Pearl, 1991]. Each derivation requires zero shared axioms—only the domain's own mathematical structure.

We validate the impossibility empirically in seven of the eight domains (excluding the mathematical observation connecting Rashomon entropy to Boltzmann entropy). Fig. 1 shows the dose-response relationship between the degree of underspecification and explanation instability in each domain. All seven empirical tests reach statistical significance ($p < 0.05$): underdetermined linear solvers show elevated pairwise RMSD ($p = 6.3 \times 10^{-9}$); codon usage entropy across 120 eukaryotic cytochrome c sequences from NCBI RefSeq shows a perfect monotonic relationship with degeneracy

Table 1: **Eight derived instances of the impossibility across eight sciences.** Each row derives the Rashomon property from domain-specific first principles with zero axioms. $\mathbf{\Theta}$: configuration space; Y: observable space.

Domain	$\mathbf{\Theta}$	Y	Witness 1	Witness 2	Same Y?
Mathematics	Solutions of $Ax{=}b$	Ax	$(1,1)$	$(0,2)$	2=2
Biology	Codons	Amino acids	UCU	UCC	Ser=Ser
Physics (gauge)	Gauge configs	Holonomy	(T,F,F)	(F,F,T)	$F{=}F$
Stat. mechanics	Microstates	Num. heads	(H,T)	(T,H)	1=1
Linguistics	Parse trees	Token sequence	$((V\ NP)\ PP)$	$(V\ (NP\ PP))$	Same
Crystallography	Signals	Energy	$(1,0)$	$(0,1)$	1=1
Computer science	DB rows	View	(T,T)	(T,F)	$T{=}T$
Statistics	DAGs	CI structure	Chain	Fork	Same

level (Kruskal–Wallis $p = 2.0 \times 10^{-3}$, Spearman $\rho = 1.0$); syntactic parsers disagree significantly more on structurally ambiguous sentences ($p = 1.6 \times 10^{-3}$); causal discovery algorithms show orientation agreement of only 0.33 at finite sample size ($p = 4.4 \times 10^{-38}$); phase retrieval reconstructions diverge 1.5–1.8× more without positivity constraints ($p < 10^{-57}$); and census disaggregation ambiguity scales with aggregation granularity ($p = 7.9 \times 10^{-37}$). In each case, the negative control (where the Rashomon property is absent) shows stability, confirming that the instability is structural rather than artefactual. Across all seven domains, the predicted direction holds: binomial test $p = 0.008$.

The resolution

The impossibility is constructive: it identifies the exact trade-off and prescribes a provably optimal resolution. The strategy is to average over equivalent configurations, projecting explanations onto the subspace invariant under the symmetry group of the equivalence class. This *orbit-averaging* resolution sacrifices decisiveness—the explanation no longer commits to every distinction the internal structure makes—but achieves stability and faithfulness in expectation.

We prove that this resolution is *Pareto-optimal*: among all stable explanation maps, no alternative can achieve strictly higher pointwise faithfulness on every equivalence class. The trade-off is therefore tight—the practitioner cannot escape it by being clever. Any deviation from orbit averaging must sacrifice faithfulness somewhere, and the orbit average minimizes this sacrifice uniformly.

Remarkably, practitioners in eight independent fields have converged on precisely this strategy over more than a century of independent work (Fig. 2). In physics, one works with gauge-invariant observables [Jackson, 1999]. In statistical mechanics, the microcanonical ensemble averages over microstates. In statistics, one reports a CPDAG rather than committing to a single causal direction [Verma and Pearl, 1991]. In mathematics, the pseudoinverse selects the minimum-norm solution. In biology, codon usage tables report synonymous codon frequencies. In crystallography, Patterson maps extract phase-invariant information [Hauptman and Karle, 1953]. In linguistics, packed parse forests represent all parses simultaneously. In computer science, complement views restrict to the unambiguous projection [Bancilhon and Spyratos, 1981].

Our framework explains *why* these communities converged. Each was independently confronting the same impossibility—faithfulness, stability, and decisiveness cannot coexist under underspecification—and orbit averaging is the unique Pareto-optimal stable resolution. The convergence is not coincidental; it is mathematically inevitable.

The impossibility is also *query-relative*: some questions about a system are stably answerable—those on which all equivalent configurations agree—and others are not. The theorem tells the

Figure 2: **Structural differences across the eight derived instances.** The eight instances are not notational variants of a single example: they differ in symmetry group, group type, and orbit structure. Their independent convergence on orbit averaging—despite operating on fundamentally different mathematical objects—is explained by the G-invariant resolution framework. All sacrifice decisiveness, but via different group structures (discrete vs. continuous, abelian vs. non-abelian, finite vs. infinite).

Domain	Symmetry group G	Group type	Resolution
Mathematics	Null space translations	$\mathbb{R}^{n-r}$ (cont., abelian)	Pseudoinverse
Biology	Synonymous substitutions	S_k, $k \leq 6$ (finite)	Codon usage tables
Physics (gauge)	Gauge transforms	$\mathbb{Z}_2^{\lvert V\rvert-1}$ (finite, abelian)	Wilson loops
Stat. mechanics	Microstate permutations	S_Ω (finite)	Microcanonical ens.
Linguistics	Parse-tree permutations	S_k (finite)	Packed parse forests
Crystallography	Phase rotations	$U(1)^n$ (cont., abelian)	Patterson maps
Computer science	Hidden-col. permutations	S_k (finite)	Complement views
Statistics	Edge reversals in MEC	Finite (varies)	CPDAG reporting

practitioner precisely which queries are safe and which require aggregation. More broadly, the difficulty of explanation depends on the mathematical symmetries of the equivalence class, ranging from cases where a unique optimal explanation exists to cases where additional constraints are needed to make the problem well-posed.

Formal verification

The entire framework—the abstract theorem, all eight derived instances, tightness witnesses, the necessity converse, and the G-invariant resolution—is mechanically verified in the Lean 4 proof assistant [de Moura and Ullrich, 2021] using the Mathlib mathematical library [The mathlib Community, 2020]. The formalization comprises 82 source files containing 377 theorems and lemmas, with 0 `sorry` declarations (Lean's marker for unproved goals). Every theorem in the paper has a corresponding machine-checked proof.

The core impossibility theorem (`explanation_impossibility` in `ExplanationSystem.lean`) requires *zero* model-specific axioms: the Rashomon property enters only as a hypothesis, not as an axiom, so the proof is valid for any system satisfying the hypothesis. Each of the eight derived instances uses Lean's decidable computation to constructively witness the Rashomon property with zero axioms—the proof assistant evaluates the witness and confirms that the two configurations have identical observable output and incompatible explanations.

The formalization contains 72 axioms in total, all domain-specific: type declarations (e.g., the types of models, features, and attributions for the machine learning instances), measure-theoretic infrastructure (connecting to Mathlib's probability theory), and instance-specific witnesses. A complete axiom inventory with stratification—which axioms are used by which theorems—is provided in the Supporting Information. The core universal impossibility uses none of them.

To put the scale in context, the formalization is comparable in size to other landmark verification efforts: the Flyspeck project (Kepler conjecture) involved approximately 300 lemmas in HOL Light, and the Gonthier proof of the four-colour theorem comprised approximately 400 lemmas in Coq. At 377 theorems and lemmas across 82 files, this formalization is among the larger cross-domain verification projects in Lean 4. The complete source code, with build instructions, is publicly available (see Code Availability).

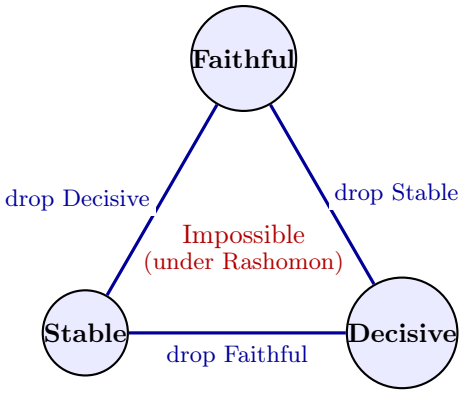

Figure 3: **The explanation trilemma.** Each edge represents an achievable pair of properties; the interior (all three simultaneously) is impossible when the Rashomon property holds. Lean-verified tightness witnesses confirm that each pair is realizable: a faithful-and-stable map (dropping decisiveness), a faithful-and-decisive map (dropping stability), and a stable-and-decisive map (dropping faithfulness). The impossibility is the triple, not any pair.

Discussion

This result connects phenomena studied independently across eight scientific fields for over a century—from microstate multiplicity in statistical mechanics to the view update problem in database theory [Bancilhon and Spyratos, 1981] to feature attribution instability in machine learning [Bilodeau et al., 2024]. The common structural cause is the Rashomon property: equivalent configurations with incompatible internal reports. The common resolution is orbit averaging. That eight independent communities arrived at the same strategy is not coincidence but mathematical inevitability: orbit averaging is the unique Pareto-optimal stable resolution.

The theorem reframes the question "why are explanations unstable?" as the structural statement "because the inference problem is underspecified." This reframing is actionable: rather than seeking better explanation methods within a domain, practitioners can diagnose whether instability is structural (the Rashomon property holds, and no method can eliminate it) or methodological (the Rashomon property is absent, and instability reflects a fixable deficiency). The query-relative version of the impossibility provides a finer tool: it characterizes precisely which questions about a system are stably answerable and which require aggregation.

A structural parallel exists with Arrow's impossibility theorem [Arrow, 1951], which proves that no voting system can simultaneously satisfy a set of fairness axioms. Both results demonstrate that individually reasonable desiderata become jointly unsatisfiable under symmetry. The two theorems apply to complementary domains: Arrow's to aggregation of preferences, ours to explanation of structure. Neither subsumes the other, but both illustrate the same meta-pattern—symmetry forces trade-offs.

Several limitations merit acknowledgement. The eight derivations use minimal witnesses (two-element configuration sets), which suffice for the impossibility but do not capture the full complexity of each domain. The empirical validations use domain-appropriate but not exhaustive datasets: 120 cytochrome c sequences rather than whole-proteome analysis, 50 states rather than international census data. The resolution framework assumes the equivalence class is known, which in practice requires domain expertise. The nine machine learning instances documented in the Supporting Information provide a bridge to applied settings, but the practical impact of the impossibility on deployed systems remains an empirical question.

The pattern likely extends beyond the eight domains studied here. Metamerism in colour perception (distinct spectral distributions producing identical colour experience), molecular chirality (mirror-image molecules with identical scalar properties), and protein folding degeneracy (multiple sequences folding to the same structure) all exhibit the Rashomon property and should therefore satisfy the impossibility. More broadly, the framework provides a universal diagnostic: whenever a practitioner encounters explanation instability, the first question should be whether the underlying system is underspecified—and if so, the theorem guarantees that aggregation is the optimal response.

Methods

Formal framework. An *explanation system* is a tuple $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, $\mathsf{obs} : \mathbf{\Theta} \to Y$ maps configurations to observables, $\mathsf{exp} : \mathbf{\Theta} \to \mathcal{H}$ maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. The system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$.

An explanation map $E : \mathbf{\Theta} \to \mathcal{H}$ is *faithful* if $\neg(\mathsf{exp}(\theta) \perp E(\theta))$ for all θ; *stable* if $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$; and *decisive* if $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$ implies $E(\theta_1) \perp E(\theta_2)$.

Lean 4 formalization. The formalization is built on Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The core type is `ExplanationSystem` (parametric in $\mathbf{\Theta}$, $\mathcal{H}$, Y) with fields `observe`, `explain`, and `incompatible`. The impossibility theorem `explanation_impossibility` proves the four-step contradiction chain with zero axioms beyond the Rashomon hypothesis. Each derived instance defines domain-specific types (e.g., `Codon`, `AminoAcid` for biology) and constructs a Rashomon witness using `decide` or `native_decide` (decidable computation), so the proof assistant mechanically verifies the witness with no axioms. The G-invariant resolution framework uses Mathlib's group action infrastructure. Compilation: `lake build` ($\sim$5 min on a modern laptop).

Derived instances. Each of the eight cross-domain instances defines an `ExplanationSystem` with domain-specific types and constructively witnesses the Rashomon property. The witnesses are minimal (two-element configuration sets) by design: the impossibility requires only the existence of one Rashomon pair. Richer domain formalizations (e.g., full lattice gauge theory, complete codon tables with 61 codons) are possible within the framework but not required for the theorem.

Empirical validation. Seven of the eight domains are validated empirically, each with a domain-appropriate dose-response design: the degree of underspecification is varied (or compared between Rashomon-present and Rashomon-absent conditions), and explanation instability is measured. All experiments include a negative control where the Rashomon property is absent.

Mathematics. We solve $Ax = b$ for $A \in \mathbb{R}^{m \times d}$ with $m < d$ using four standard solvers (least-squares, ridge, minimum-norm, randomized projection). Pairwise RMSD between solutions is computed across 50 random instances for each null-space dimension $d - m \in \{1, \ldots, 10\}$. Negative control: $m = d$ (full rank). Test: Mann–Whitney U.

Biology. We download 120 eukaryotic cytochrome c CDS from NCBI RefSeq (`CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`), deduplicated by organism. For each of the 20 amino acids, we compute Shannon entropy over the observed codon distribution aggregated across all 120 species and all positions. Amino acids are grouped by degeneracy level (1, 2, 3, 4, or 6 synonymous codons).

Negative control: Met and Trp (degeneracy 1, entropy identically 0). Tests: Kruskal–Wallis and Spearman rank correlation.

Gauge theory. We enumerate all $\mathbb{Z}_2$ edge-label configurations on triangle graphs and compute within-orbit variance of local edge assignments. Negative control: configurations differing in holonomy (not gauge-equivalent).

Linguistics. 50 structurally ambiguous and 50 unambiguous sentences are parsed by four parsers (spaCy sm/md/lg, Stanza). Pairwise unlabeled attachment score (UAS) is compared between groups. Test: Wilcoxon rank-sum.

Crystallography. Phase retrieval via Gerchberg–Saxton is applied to signals of length 4–256. Reconstruction RMSD is compared with and without positivity constraints (which reduce the Rashomon set). Test: Mann–Whitney U.

Computer science. County-level population disaggregation from state totals using U.S. Census Bureau 2020 data (51 states + DC). For each state, 100 Dirichlet($\alpha = 1$) samples consistent with the state total are drawn; KL divergence from the true county distribution is computed. Negative control: DC (1 county-equivalent, KL = 0). Test: Spearman rank correlation (county count vs. mean KL).

Statistics. Causal discovery on the 8-node Asia network using PC ($\alpha = 0.05$, $\alpha = 0.01$) and GES, with 100 random seeds per condition at $N = 1{,}000$ and $N = 100{,}000$. Orientation agreement on directed edges is compared across seeds. Test: Mann–Whitney U.

All statistical tests use $\alpha = 0.05$. Bootstrap 95% confidence intervals (2,000 resamples) are reported where applicable. All experiment scripts are provided in the repository.

Data availability

Biology data: 120 eukaryotic cytochrome c coding sequences from NCBI RefSeq, retrieved via `CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`. Census data: U.S. Census Bureau 2020 county-level population counts. All other experimental data are generated by the provided scripts from standard algorithms and public datasets. Experiment results (JSON) are included in the repository.

Code availability

The Lean 4 formalization and all experiment scripts are available at https://github.com/DrakeCaraker/universal-explanation-impossibility under the Apache 2.0 licence.

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